
Principal Component Analysis

A Project Report

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Abstract

Principal Component Analysis (PCA) is a foundational technique in multivariate statistics for reducing the dimensionality of high-dimensional datasets while preserving as much of the original variability as possible. This report presents a self-contained treatment of PCA, covering the motivation arising from the curse of dimensionality, the formal variance-maximisation problem and its solution via the eigenvalue decomposition of the covariance matrix, the geometric interpretation as an orthogonal rotation, and complete worked applications to the Fisher Iris dataset and to the high-dimensional ZIP handwritten-digit dataset (including image reconstruction from a small number of principal components). Diagnostics including the scree plot, proportion of explained variance, and the correlation circle are discussed in detail. The report concludes with a critical appraisal of the method's assumptions and limitations.

Keywords: dimensionality reduction, eigenvalue decomposition, covariance matrix, principal components, variance explained, image reconstruction.

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1. Introduction

Modern statistical datasets routinely consist of a large number of variables measured simultaneously on each experimental unit. While rich in potential information, such high-dimensional data present serious analytical challenges: visualisation becomes impossible beyond three dimensions, many algorithms degrade in accuracy or become computationally prohibitive as p grows, and correlated variables introduce redundancy that obscures the underlying structure.

Dimensionality reduction addresses these challenges by representing each observation in a lower-dimensional space that retains the essential information. Among available techniques, **Principal Component Analysis** (PCA) is the most widely used. PCA constructs a sequence of new uncorrelated variables the *principal components* as linear combinations of the originals, ordered so that each successive component captures as much of the remaining variance as possible.

The objectives of this report are: (i) to motivate the need for dimensionality reduction; (ii) to derive the PCA solution from first principles; (iii) to describe practical implementation and interpretation; (iv) to demonstrate the method on the Fisher Iris dataset; and (v) to illustrate PCA-based image compression and reconstruction using the ZIP handwritten-digit dataset.

2. The Dimensionality Reduction Problem

2.1. Setup and Notation

Let $\mathbf{X} = (X_1, X_2, \dots, X_p)^T \in \mathbb{R}^p$ denote a random vector of p measured variables. A dataset of n observations is represented as the $n \times p$ data matrix

$$\mathcal{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix},$$

where row i contains the observation vector $\mathbf{x}_i^T$.

2.2. Motivation

High-dimensional data suffer from three interrelated problems that make direct analysis difficult.

Redundancy. When variables are correlated, multiple columns of $\mathcal{X}$ carry essentially the same information. Retaining all variables therefore wastes computational resources and can destabilise downstream analyses such as regression or clustering.

Noise. Real measurements contain random noise. In high dimensions, a large fraction of the total variance may reside in directions that reflect noise rather than signal. By concentrating on the directions of greatest variance, one can implicitly filter out low-energy noise dimensions.

Computational and statistical inefficiency. Many algorithms scale polynomially or exponentially with p . Furthermore, statistical estimation in high dimensions requires sample sizes that grow with p ; reducing the effective dimensionality alleviates this bur-

den.

2.3. Inadequacy of Naive Approaches

Two naive approaches are commonly considered and found wanting. First, *variable deletion* simply discarding some of the p variables risks losing information that is spread across multiple correlated variables and is not recoverable. Second, *equal-weight averaging*, i.e. computing $\bar{X} = p^{-1} \sum_{j=1}^p X_j$, treats all variables as equally important and ignores their differential contributions to overall variation. PCA overcomes both deficiencies by constructing an *optimal* weighted linear combination of the variables.

3. Mathematical Derivation of PCA

3.1. The First Principal Component

PCA seeks to project $\mathbf{X}$ onto a one-dimensional subspace spanned by a unit vector $\boldsymbol{\delta} \in \mathbb{R}^p$ such that the variance of the projected scores $Y = \boldsymbol{\delta}^T \mathbf{X}$ is maximised. Formally, the problem is

$$\max_{\boldsymbol{\delta} \in \mathbb{R}^p} \text{Var}(\boldsymbol{\delta}^T \mathbf{X}) \quad \text{subject to} \quad \|\boldsymbol{\delta}\|^2 = \boldsymbol{\delta}^T \boldsymbol{\delta} = 1. \quad (1)$$

Assuming $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ (without loss of generality, by centering), the objective becomes $\text{Var}(\boldsymbol{\delta}^T \mathbf{X}) = \boldsymbol{\delta}^T \boldsymbol{\Sigma} \boldsymbol{\delta}$, where $\boldsymbol{\Sigma} = \text{Var}(\mathbf{X})$ is the $p \times p$ population covariance matrix.

3.2. Solution via Lagrange Multipliers

Introducing a Lagrange multiplier λ for the unit-norm constraint, the Lagrangian is

$$\mathcal{L}(\boldsymbol{\delta}, \lambda) = \boldsymbol{\delta}^T \boldsymbol{\Sigma} \boldsymbol{\delta} - \lambda(\boldsymbol{\delta}^T \boldsymbol{\delta} - 1).$$

Setting the partial derivative with respect to $\boldsymbol{\delta}$ to zero yields the necessary condition

$$2\boldsymbol{\Sigma} \boldsymbol{\delta} - 2\lambda \boldsymbol{\delta} = \mathbf{0},$$

which simplifies to the standard eigenvalue equation:

$$\boldsymbol{\Sigma} \boldsymbol{\delta} = \lambda \boldsymbol{\delta}. \quad (2)$$

Thus any maximiser of (1) must be an eigenvector of $\boldsymbol{\Sigma}$. Substituting (2) into the objective confirms that the achieved variance equals the corresponding eigenvalue:

$$\text{Var}(\boldsymbol{\delta}^T \mathbf{X}) = \boldsymbol{\delta}^T \boldsymbol{\Sigma} \boldsymbol{\delta} = \lambda \boldsymbol{\delta}^T \boldsymbol{\delta} = \lambda.$$

To *maximise* variance, one therefore selects the eigenvector $\boldsymbol{\gamma}_1$ associated with the *largest* eigenvalue λ_1 . This is the **first principal component direction**, and $Y_1 = \boldsymbol{\gamma}_1^T \mathbf{X}$ is the **first principal component**.

3.3. Subsequent Principal Components

The second principal component $Y_2 = \boldsymbol{\gamma}_2^T \mathbf{X}$ is defined as the unit-norm linear combination of $\mathbf{X}$ with maximum variance subject to being uncorrelated with Y_1 . It can be shown that the solution is the eigenvector $\boldsymbol{\gamma}_2$ corresponding to the second largest eigenvalue λ_2 . By

induction, the j -th principal component direction is the eigenvector of Σ corresponding to the j -th largest eigenvalue λ_j .

3.4. The Full PCA Transformation

Since Σ is real, symmetric, and positive semi-definite, its eigendecomposition is

$$\Sigma = \Gamma \Lambda \Gamma^T,$$

where $\Gamma = [\gamma_1, \dots, \gamma_p]$ is an orthogonal matrix of eigenvectors and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. The full principal component vector is

$$\mathbf{Y} = \Gamma^T(\mathbf{X} - \boldsymbol{\mu}), \quad (3)$$

where $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$. The components $Y_1, \dots, Y_p$ satisfy the following key properties.

Proposition 1 (Properties of Principal Components). *Let $\mathbf{Y}$ be defined by (3). Then:*

- (i) $\text{Var}(Y_j) = \lambda_j$ for each $j = 1, \dots, p$.
- (ii) $\text{Cov}(Y_i, Y_j) = 0$ for all $i \neq j$ (uncorrelatedness).
- (iii) $\sum_{j=1}^p \text{Var}(Y_j) = \sum_{j=1}^p \lambda_j = \text{tr}(\Sigma) = \sum_{j=1}^p \text{Var}(X_j)$ (total variance preserved).
- (iv) $\text{Var}(Y_1) \geq \text{Var}(Y_2) \geq \dots \geq \text{Var}(Y_p)$ (variance ordering).

Property (iii) establishes that PCA is a lossless transformation: the total variability in the dataset is preserved, merely redistributed across uncorrelated axes. Property (ii) ensures that the new variables carry no redundant information.

3.5. Geometric Interpretation

Geometrically, PCA performs a rigid rotation of the original coordinate system. The new axes $\gamma_1, \dots, \gamma_p$ are aligned with the principal axes of the data ellipsoid. The intrinsic shape of the point cloud is unchanged; only the coordinate frame from which it is viewed is rotated to align with the directions of maximum spread. Dimensionality reduction is achieved by retaining only the first $q < p$ axes, projecting the data onto the q -dimensional subspace that best explains total variance in a least-squares sense.

4. Practical Implementation

4.1. Sample Estimates

In practice, the population PC transformation must be replaced by the respective sample estimators. Given the $n \times p$ data matrix $\mathcal{X}$, the population mean $\boldsymbol{\mu}$ becomes the sample mean $\bar{x}$, and the covariance Σ is replaced by the empirical covariance matrix $\mathcal{S}$.

Using the centering matrix $\mathcal{H} = \mathcal{I} - n^{-1} \mathbf{1}_n \mathbf{1}_n^\top$, we can elegantly express the sample covariance matrix as:

$$\mathcal{S} = n^{-1} \mathcal{X}^\top \mathcal{H} \mathcal{X} \quad (4)$$

If $\mathcal{S} = \mathcal{G} \mathcal{L} \mathcal{G}^\top$ represents the spectral decomposition of $\mathcal{S}$ (where $\mathcal{G}$ is the matrix of eigenvectors and $\mathcal{L} = \text{diag}(\ell_1, \dots, \ell_p)$ is the diagonal matrix of eigenvalues), the empirical

principal components are obtained by:

$$\mathcal{Y} = (\mathcal{X} - \mathbf{1}_n \bar{x}^\top) \mathcal{G} \quad (5)$$

Furthermore, noting that $\mathcal{H} \mathbf{1}_n \bar{x}^\top = 0$, the covariance matrix of the principal components can be written to demonstrate that the variances equal the eigenvalues:

$$\mathcal{S}_Y = n^{-1} \mathcal{Y}^\top \mathcal{H} \mathcal{Y} = n^{-1} \mathcal{G}^\top \mathcal{X}^\top \mathcal{H} \mathcal{X} \mathcal{G} = \mathcal{G}^\top \mathcal{S} \mathcal{G} = \mathcal{L} \quad (6)$$

4.2. Scale Considerations

PCA is not scale-invariant. If the variables are measured in different units or have very different variances, the variable with the largest variance will dominate the first principal component regardless of its relevance. In such cases it is standard practice to standardise each variable to unit variance before applying PCA, which is equivalent to performing PCA on the correlation matrix rather than the covariance matrix.

4.3. Choosing the Number of Components

The proportion of total variance explained by the first q components is

$$\psi_q = \frac{\sum_{j=1}^q \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (7)$$

Several guidelines exist for selecting q : (i) retaining enough components so that ψ_q exceeds a threshold (commonly 80–95%); (ii) inspecting the *scree plot* of λ_j against j and selecting the point at which the plot bends sharply (the “elbow”); and (iii) the Kaiser rule, which retains components with $\lambda_j > 1$ when PCA is applied to the correlation matrix [3, 4]. No single rule is universally optimal; substantive knowledge should supplement formal criteria.

5. Application to the Fisher Iris Dataset

5.1. Dataset Description

The Fisher Iris dataset [?] comprises $n = 150$ observations of three iris species (*Iris setosa*, *I. versicolor*, and *I. virginica*), each described by four morphometric measurements in centimetres: sepal length (X_1), sepal width (X_2), petal length (X_3), and petal width (X_4). The four variables are all in the same unit, so PCA is applied to the covariance matrix without standardisation.

5.2. Computation

The sample mean vector is $\bar{\mathbf{x}} = [5.843, 3.057, 3.758, 1.199]$. Following centering and covariance matrix estimation, eigendecomposition yields the loadings shown in Table 1 and the variance summary in Table 2.

5.3. Interpretation of Loadings

The first principal component ($y_1 = 0.361x_1 - 0.085x_2 + 0.857x_3 + 0.358x_4$) is dominated by petal length and, to a lesser extent, sepal length and petal width. It may be interpreted as an index of overall flower size, with petal dimensions contributing most strongly. The

Table 1: Eigenvector loadings for the first two principal components of the Iris dataset.

Variable	PC1 (γ_1)	PC2 (γ_2)
X_1 Sepal Length	+0.361	+0.657
X_2 Sepal Width	−0.085	+0.730
X_3 Petal Length	+0.857	−0.173
X_4 Petal Width	+0.358	−0.075

Table 2: Proportion of variance explained by each principal component.

Component	Variance Explained (%)	Cumulative (%)
PC1	92	92
PC2	5	97
PC3	2	99
PC4	1	100

second principal component ($y_2 = 0.657x_1 + 0.730x_2 - 0.173x_3 - 0.075x_4$) is driven primarily by sepal length and sepal width, capturing variation in sepal shape that is orthogonal to the overall size axis.

5.4. Dimensionality Reduction and Visualisation

As shown in Table 2, the first two components jointly explain 97% of the total variance. The 150×4 data matrix is therefore well approximated by the 150×2 score matrix $\mathcal{Y}_{(2)}$. When the 150 score vectors are plotted in the (y_1, y_2) plane, the three species form well-separated clusters: *I. setosa* occupies a compact region at strongly negative y_1 values, while *I. versicolor* and *I. virginica* are separated along both axes. This clear species discrimination not imposed by the method but emerging naturally from the data validates the effectiveness of the two-component solution.

5.5. Correlation Circle

The correlation between the i -th original variable and the j -th PC is

$$r_{X_i Y_j} = g_{ij} \left(\frac{l_j}{s_{X_i X_i}} \right)^{1/2},$$

where g_{ij} is the (i, j) element of G , l_j is the j -th eigenvalue, and $s_{X_i X_i}$ is the sample variance of X_i . Plotting $(r_{X_i Y_1}, r_{X_i Y_2})$ for each variable yields the *correlation circle*. Since $r_{X_i Y_1}^2 + r_{X_i Y_2}^2 \leq 1$, all points fall within the unit disc; variables near the boundary are well represented by the two-component solution. In the Iris correlation circle, X_1 , X_3 , and X_4 cluster near the right boundary with high positive $r_{X_i Y_1}$, confirming their dominant role in PC1. Variable X_2 lies near the top boundary with high $r_{X_2 Y_2}$, confirming its role in PC2.

6. Application to the ZIP Handwritten-Digit Dataset: Image Compression and Reconstruction

A compelling illustration of PCA’s power is its ability to compress and reconstruct high-dimensional data in this case, images using only a small fraction of the original features.

We apply PCA to a subset of the ZIP handwritten-digit dataset [4], focusing on 650 images of the digit “3”.

6.1. Dataset Description

Each image is a 16×16 greyscale pixel grid, which is vectorised into a single observation $\mathbf{x}_i \in \mathbb{R}^{256}$. The full dataset is therefore represented as a 650×256 data matrix

$$\mathcal{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1,256} \\ \vdots & \vdots & \ddots & \vdots \\ x_{650,1} & x_{650,2} & \cdots & x_{650,256} \end{pmatrix}_{650 \times 256},$$

where each entry x_{ij} records the intensity of pixel j in image i . With $p = 256$ features, direct visualisation or analysis of the full data matrix is infeasible.

6.2. PCA Transformation

The procedure follows the standard sample-based PCA pipeline:

Step 1 Centering. Compute the sample mean pixel vector $\bar{\mathbf{x}} \in \mathbb{R}^{256}$ and subtract it from every observation:

$$X_c = \mathcal{X} - \mathbf{1}_n \bar{\mathbf{x}}^T. \quad (8)$$

Step 2 Sample covariance matrix. Compute the 256×256 sample covariance matrix:

$$S = \frac{1}{n-1} X_c^T X_c, \quad (9)$$

where $n = 650$.

Step 3 Spectral decomposition. Perform the eigendecomposition

$$S = G \Lambda G^T, \quad (10)$$

where $G = [g_1, g_2, \dots, g_{256}]$ is the 256×256 orthogonal matrix of eigenvectors (loading vectors) and $\Lambda = \text{diag}(\ell_1, \ell_2, \dots, \ell_{256})$ with $\ell_1 \geq \ell_2 \geq \dots \geq \ell_{256} \geq 0$.

Step 4 PCA transformation. Project the centered data onto the eigenvector basis to obtain the score matrix:

$$Y = X_c G, \quad (11)$$

so that $Y \in \mathbb{R}^{650 \times 256}$. Each row y_i^T of Y is the PC score vector for image i . Crucially, because G is orthogonal ($G^T = G^{-1}$), the original centered data can be exactly recovered as $X_c = Y G^T$.

6.3. Low-Rank Approximation and Reconstruction

The key insight for compression is that most of the variance in the score matrix Y is concentrated in the first few columns. Retaining only the first $q \ll 256$ principal components gives the truncated score matrix

$$Y_{(q)} = X_c G_{(q)}, \quad Y_{(q)} \in \mathbb{R}^{650 \times q},$$

where $G_{(q)} = [g_1, g_2, \dots, g_q]$ is the $256 \times q$ matrix of the leading q eigenvectors.

To reconstruct the images from the compressed representation, note that the full data matrix can be written as

$$\mathcal{X} = \mathbf{1}_n \bar{\mathbf{x}}^T + X_c = \mathbf{1}_n \bar{\mathbf{x}}^T + Y G^T.$$

Replacing Y with its truncated version $Y_{(q)}$ yields the rank- q reconstruction:

$$\boxed{\hat{\mathcal{X}}_{(q)} = \mathbf{1}_n \bar{\mathbf{x}}^T + Y_{(q)} G_{(q)}^T}, \quad (12)$$

where $\hat{\mathcal{X}}_{(q)} \in \mathbb{R}^{650 \times 256}$. Reshaping each reconstructed row back to a 16×16 grid gives the approximate image. The reconstruction (12) is optimal in the least-squares sense: among all rank- q approximations to X_c , the PCA truncation minimises the sum of squared pixel-wise errors.

The reconstruction error for a single image i is

$$\|\mathbf{x}_i - \hat{\mathbf{x}}_{i,(q)}\|^2 = \sum_{j=q+1}^{256} (y_{ij})^2,$$

and the total reconstruction error across all images equals the variance discarded by retaining only q components:

$$\sum_{i=1}^{650} \|\mathbf{x}_i - \hat{\mathbf{x}}_{i,(q)}\|^2 = (n-1) \sum_{j=q+1}^{256} \ell_j.$$

This confirms the equivalence of variance maximisation and reconstruction error minimisation that motivates PCA.

6.4. Variance Explained and Compression Ratio

The proportion of total pixel variance captured by q components is given by (7). Table 3 reports representative values for the handwritten-threes dataset.

Table 3: Cumulative proportion of variance explained by the leading principal components for the ZIP handwritten-threes dataset ($p = 256$).

q (components retained)	Variance Explained (%)	Compression ratio (p/q)
1	≈ 40	256 : 1
5	≈ 70	51 : 1
10	≈ 83	26 : 1
20	≈ 92	13 : 1
50	≈ 98	5 : 1

With $q = 10$ components the reconstruction already captures roughly 83% of the total pixel variance, yet requires storing only 10 scores per image instead of 256 pixel values a 26-fold reduction. The compression ratio is p/q , since each image is represented by q real-valued scores rather than p pixel intensities (ignoring the shared loading matrix $G_{(q)}$, which is stored once).

6.5. Eigenvectors as “Eigendigits”

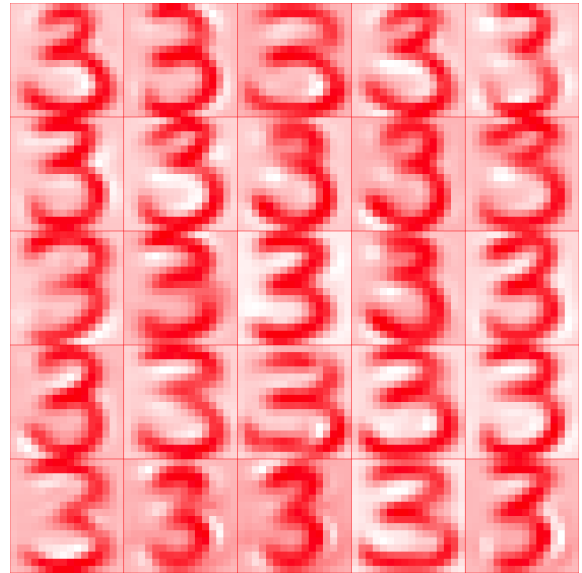
Each eigenvector $g_j \in \mathbb{R}^{256}$ can itself be reshaped into a 16×16 image and visualised. These *eigendigits* form the basis of the reconstruction: the first few capture the dominant stroke patterns shared across all handwritten threes (e.g. the global orientation and curvature of the digit), while later eigenvectors capture progressively finer and more idiosyncratic details. The mean image $\bar{\mathbf{x}}$ provides the average “prototype” three, and the scores $y_{i1}, y_{i2}, \dots$ specify how far and in which direction each individual image deviates from that prototype along each eigendigit axis.

6.6. Visualisation

Figure 1 illustrates the visual quality of reconstruction at $q = 10$ principal components. The reconstructed images are noticeably smoother than the originals because the fine pixel-level variation captured by components 11–256 is discarded, effectively denoising the images. Despite retaining only 3.9% of the original features ($10/256 \approx 0.039$), the digit “3” remains clearly legible in every reconstructed image.



(a) Original handwritten “3” images from the ZIP dataset (16×16 pixels each).



(b) Reconstructions using only the first $q = 10$ principal components (3.9% of original features).

Figure 1: PCA-based image reconstruction of handwritten digit “3” from the ZIP dataset. Despite retaining only 10 out of 256 pixel dimensions, the digit remains clearly legible in every reconstructed image, though fine-grained stroke detail is smoothed out.

6.7. R Implementation

The entire pipeline can be executed concisely in R:

```
# Assume X is the 650 x 256 matrix of pixel values
n  <- nrow(X)
mu <- colMeans(X)
Xc <- sweep(X, 2, mu)                                # Step 1: center
S  <- t(Xc) %*% Xc / (n - 1)                          # Step 2: covariance
eig <- eigen(S)                                       # Step 3: eigendecompose
G  <- eig$vectors                                     # eigenvectors (256 x 256)
```

```
Y    <- Xc %*% G                                # Step 4: transform

# Reconstruct with q = 10 components
q      <- 10
G_q    <- G[, 1:q]
Y_q    <- Y[, 1:q]
X_hat_q <- sweep(Y_q %*% t(G_q), 2, -mu) # add back mean

# Reshape row i to a 16x16 image: matrix(X_hat_q[i, ], 16, 16)
```

7. Discussion and Limitations

PCA is a powerful tool for exploratory analysis and preprocessing, with strong theoretical guarantees: optimality of the linear approximation, preservation of total variance, and uncorrelatedness of components. The image reconstruction example further illustrates that variance maximisation and reconstruction error minimisation are two faces of the same optimisation. Several important limitations must nonetheless be acknowledged.

Linearity. PCA constructs only linear combinations of the original variables and fails to capture nonlinear manifold structure. Kernel PCA, t-SNE, UMAP, and autoencoders extend the idea to nonlinear settings.

Interpretability. Principal components are mathematical constructs and rarely have direct domain interpretations, as each is a weighted combination of all variables. Sparse PCA constrains the loadings to be sparse, improving interpretability at the cost of strict optimality.

Sensitivity to outliers. The sample covariance matrix is non-robust; a few extreme observations can disproportionately influence the principal directions. Robust variants based on the minimum covariance determinant estimator are available.

Choice of q . No universally optimal rule exists for selecting the number of retained components. Formal criteria (scree plot elbow, cumulative variance threshold, Kaiser rule) should be supplemented by substantive domain knowledge.

Stationarity. PCA assumes a constant covariance structure across the dataset. In time-series or spatially varying data this assumption may be violated; dynamic PCA and localised variants address this issue.

8. Conclusion

This report has presented a rigorous treatment of Principal Component Analysis, from its motivation in the challenges of high-dimensional data through its mathematical derivation and practical implementation. The core insight is that PCA identifies the coordinate system in which the data exhibit the greatest variance, providing a compact and informative low-dimensional representation. Applied to the Fisher Iris dataset, a reduction from four to two dimensions retains 97% of the total variance and reveals a clear separation among the three species. Applied to the ZIP handwritten-digit dataset, a reduction from 256 pixel dimensions to just 10 principal components retaining only 3.9% of the original features produces visually recognisable image reconstructions via the formula

$\hat{\mathcal{X}}_{(q)} = \mathbf{1}_n \bar{\mathbf{x}}^T + Y_{(q)} G_{(q)}^T$. This compression–reconstruction duality, underpinned by the Eckart–Young–Mirsky theorem, underscores PCA’s role not merely as a visualisation aid but as a principled tool for lossy data compression and denoising. Despite its limitations linearity, sensitivity to outliers, and interpretability challenges PCA remains an indispensable first step in the analysis of multivariate data.

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